



Topic: Designing Power BI Dashboards

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Scenario Selected: End-to-End Data Analytics Project (HR Analytics – Employee Attrition)



1. Objective (Executive HR Dashboard)

For this project, I designed an **HR Executive Dashboard** to support decision-making within an organization.

The main objective of this dashboard is to help HR leaders and executives understand workforce trends, employee performance, and attrition patterns in order to improve employee retention and workforce planning.

Specifically, I aimed to answer the following business questions:

- What is the current state of the workforce?
- What is the attrition rate across departments?
- Which job roles and departments are most affected by employee turnover?
- What factors influence employee retention and attrition?

The final outcome is a **single-page interactive Power BI dashboard** designed for executive-level decision-making.

2. Dataset Setup

I used a structured HR dataset in Excel format containing employee records.

The dataset included:

- Employee demographics (age, gender, education)
- Job information (department, role, job level)
- Compensation details (monthly income, salary band)
- Work experience (years at company, total experience)
- Attrition status (Yes/No)

This dataset provided a strong foundation for analyzing workforce trends and identifying key HR issues.

3. Data Preparation

I used Power Query in Power BI to clean and prepare the data for analysis.



My key data preparation steps included:

- Removing duplicate employee records
- Handling missing values in salary and demographic fields
- Standardizing text values (e.g., department and job role names)
- Correcting data types (numeric, date, and categorical fields)
- Filtering irrelevant or inconsistent entries

This ensured that the dataset was clean, structured, and ready for modeling.

4. Data Modelling (Star Schema Design)

I designed a star schema model to improve performance and simplify analysis.

Fact Table:

- Employee Fact Table (contains attrition, salary, tenure, performance metrics)

Dimension Tables:

- Department Dimension
- Job Role Dimension
- Employee Demographics Dimension
- Time Dimension (if applicable for trend analysis)

I created relationships using a unique Employee ID, ensuring accurate filtering across all visuals.

This structure improved both performance and scalability of the dashboard.

5. Data Processing

In the Power Query editor, I created a new column called "**AttritionCount**" using a conditional column feature in Add Column. This was based on the condition:

- If Attrition = "Yes" → 1
- Else → 0



This new column was further used for creating different KPIs and charts. I then created the Attrition Rate using DAX:

$$\text{Attrition Rate} = \text{SUM}([\text{AttritionCount}]) / \text{SUM}([\text{EmployeeCount}])$$

This measure was formatted in percentage.

6. Data Analysis

Analysis involved the creation of a range of visual representations, including bar charts, key performance indicators (KPIs), table charts, pie charts, and other relevant visualizations.

These tools were utilized to gain insights and present data in a comprehensive and easily understandable manner.

Key Questions of the Dashboard:

- What is the Total Employee Count?
- What is the employee's Average Age & Average Salary?
- What is the Attrition Count of men and women?
- What are employees' working years at the company?
- Which department has maximum employees?
- What is the gender distribution?
- Which education field has maximum employees?
- Which business travel type has maximum employees?

Learned About:

- Calculated Field: To calculate Attrition Rate & Active Employees
- Matrix Table: To show Job Satisfaction rating
- Donut Chart and Pie Chart
- Bar Chart and Cluster Chart 
- KPI (Key Performance Indicators) and Slicer
- Filters: Used to filter data according to different education fields

7. Dashboard Development (Single-Page Design)



I built a single-page executive dashboard focused on clarity and storytelling.

Key KPIs Included:

- Total Employees
- Attrition Count
- Attrition Rate (%)
- Average Monthly Income
- Average Tenure

Visualizations Used:

- Bar Chart: Attrition by Department
- Pie Chart: Attrition by Gender
- Line Chart: Attrition Trend by Years at Company
- Heatmap: Job Role vs Attrition

Interactivity Features:

- Slicers for Department, Job Role, Gender, and Education Level
- Cross-filtering between visuals
- Drill-down capability for deeper analysis

The dashboard was designed to allow executives to quickly understand workforce performance at a glance.

8. Advanced Features Used

To enhance analytical depth, I implemented DAX calculations, including:

- Attrition Rate Calculation

Attrition Count = CALCULATE(COUNTROWS('Employee'), 'Employee'[Attrition] = Yes")

- Average Monthly Income



- Employee Count Measures

I also applied time-based analysis (where applicable) to observe trends in employee attrition over tenure. These advanced features improved the analytical capability of the dashboard beyond basic reporting.

9. Insights & Storytelling

Key Insights Summary:

- **Total Employees:** The organization has grown significantly, currently employing 1470 individuals, indicating substantial growth and scale.
- **Attrition Analysis:** A total of 237 employees left the organization. Among them, 150 were male, and 87 were female, indicating a higher attrition rate among males.
- **Departmental Attrition:** The Research and Development Department experienced the highest attrition rate at 56.13%, suggesting potential areas for improvement in employee retention strategies in this department.
- **Education Field Impact:** Employees in the life sciences field had the highest attrition rate, emphasizing the need to address retention challenges in this specific area.
- **Job Role Affected:** The sales role had the highest attrition rate, indicating a need for focused retention efforts in this department to reduce turnover.
- **Education-wise Attrition:** The attrition rate of High School is 18.24%, which is the highest among other education levels.
- **Attrition Rate by Gender for Different Age Group:** The attrition count among the age group of 25–34 years is 112, which is the highest among other age groups.

What should be done?

Based on these findings, I recommend:

- Reviewing salary structures for low-income employees
- Improving onboarding and early engagement programs
- Strengthening retention strategies in high-risk departments
- Providing clear career growth opportunities
- Conducting regular employee satisfaction surveys



This storytelling approach ensures the dashboard is not just descriptive but also actionable.

10. Sharing and Management

After completing the dashboard, I published it to Power BI Service.

I demonstrated how the dashboard can be shared securely with stakeholders using:

- Workspace access control
- Role-based permissions
- Secure link sharing

This ensures that decision-makers can access the dashboard anytime, from anywhere, while maintaining data security.